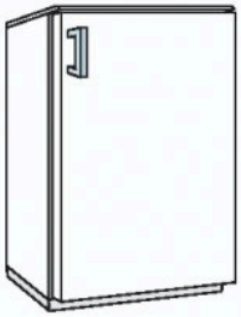
8. Say if each of the following is true or false:

(i) $9 > 0$ (ii) $-1 < 4$ (iii) $2 < -3$ (iv) $-4 > -1$
(v) $0 > -4$ (vi) $-12 > 6$ (vii) $-1 < -7$ (viii) $6 > -7$.

9. The temperature of the fridge compartment of a fridge-freezer is set at 5°C.
The freezer compartment is set at −18°C.
What is the difference between these temperature settings?



10. The table gives the highest and lowest temperatures recorded in several cities during one year.

	New York	Paris	Rome	Moscow	Sydney
Highest temperature	27°C	32°C	34°C	28°C	34°C
Lowest temperature	−9°C	−6°C	8°C	−21°C	7°C

(i) Which city recorded the lowest temperature?
(ii) Which city recorded the biggest difference between its highest and lowest temperatures?
(iii) Which city recorded the smallest difference between its highest and lowest temperatures?

11. Copy and complete the following by putting the symbols > or < in the box.

(i) $2 \square 5$ (ii) $2 \square -3$ (iii) $0 \square 4$ (iv) $0 \square -2$
(v) $-3 \square 4$ (vi) $-3 \square -4$ (vii) $-5 \square -1$ (viii) $-3 \square 0$

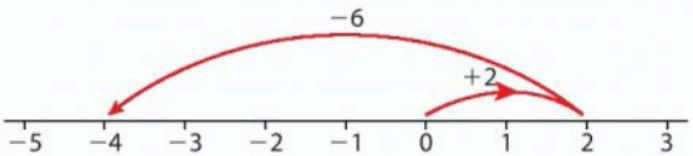
Section 2.2 Adding and subtracting integers

If the temperature is 2°C and drops 6°C, it is then −4°C.
That is, $2 - 6 = -4$.

When combining negative and positive numbers such as $2 - 6$, it can be very helpful to use the number line.

To find $2 - 6$ on the number line, start at 0, move 2 places to the right and then move 6 places to the left.

You finish at −4.



To find the value of $-1 - 4$, start at 0, move 1 place to the left and then move 4 places to the left.

The result is -5, as shown below.

Rules for adding integers

When there are too many numbers, it may be no longer convenient to use a number line to add integers.

Take these numbers: $4 - 6 + 8 - 3$

Write the positive numbers first: $4 + 8 - 6 - 3$

Now combine the positive numbers and negative numbers separately.

$$4 + 8 - 6 - 3 = 12 - 9$$
$$= 3$$

Rules for adding integers

1. When the signs are the same, add the numbers and retain the sign.
2. When the signs are different, subtract the numbers (start with the numerically bigger number) and keep the sign of the numerically bigger number.

$4 + 8 = 12$
$4 - 8 = -4$
$-4 + 8 = 4$
$-4 - 8 = -12$

Example 1

Express each of these as a single integer:

(i) $-4 - 8$ (ii) $-7 + 2$ (iii) $-2 + 5 - 13 + 6$

(i) $-4 - 8 = -12 \ldots$ retain the sign and add the numbers

(ii) $-7 + 2 = -5 \ldots$ retain sign of bigger number and take 2 from 7

(iii) $-2 + 5 - 13 + 6$

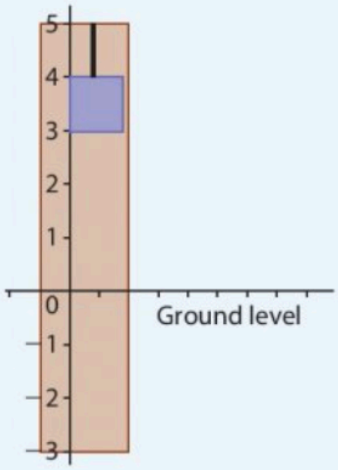
We rewrite this as $-2 - 13 + 5 + 6$

$= -15 + 11$

$= -4$

Remember; as with the scale on a thermometer, a number line can be drawn vertically as well as horizontally as the following investigation will show.

Investigation:



A lift in an hotel travels between a top floor +4 and an underground carpark at −3.

1. If the lift starts at ground level goes up two floors, down three floors, up one floor and down two floors, what level does it finish at? Represent the movement as an integer sum.
2. The lift starts at ground level as 4 people get in, goes up three floors where 3 people leave and 4 people get in, goes down five levels where 4 people leave as 5 people get in and then goes up three levels. Using integers investigate: (i) which level the lift is at now and
   (ii) how many people are still in the lift?

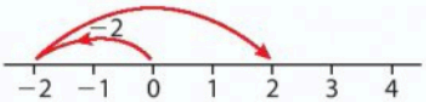
Exercise 2.2

1. Work out the following calculations using the number line.

(i) $3 - 5$ (ii) $2 - 6$ (iii) $0 - 3$ (iv) $1 - 4$
(v) $-1 + 4$ (vi) $-2 + 7$ (vii) $-1 - 3$ (viii) $-2 - 4$

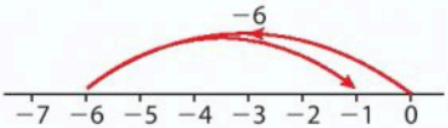
2. Write down the calculation represented by each diagram:

(i)

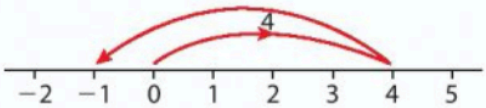


(ii)

(iii)



(iv)



3. Express each of the following as a single number.

(i) $3 + 7$ (ii) $6 - 4$ (iii) $12 - 5$ (iv) $-8 + 6$
(v) $-10 + 3$ (vi) $2 - 8$ (vii) $-12 + 7$ (viii) $-16 + 12$

4. Simplify each of these:

(i) $-1 - 2$ (ii) $-9 + 3$ (iii) $-4 - 2$ (iv) $1 - 8$
(v) $-7 - 4$ (vi) $-2 + 12$ (vii) $-12 - 3$ (viii) $-20 - 30$